

Complex Numbers

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INTRODUCTION

Complex numbers will be a fundamental part of the toolkit for this course. Using the complex roots of polynomials and allowing exponentials with complex exponents will simplify and unify our study of constant coefficient linear ODE's. The key will be Euler's formula

$$e^{j\theta} = \cos \theta + j \sin \theta$$

This formula will allow us to replace trigonometric functions which have hard-to-remember, hard-to-manipulate identities by complex exponentials which have easy-to-remember and easy-to-manipulate rules.

There are several applications of complex numbers in science and engineering, in particular in electrical alternating current theory, mechanical vector analysis, heat conduction and fluid flow.

One of the most important applications of complex numbers in engineering is in the study of alternating current (a.c.) circuits. The phase relationships between different components in a circuit can be expressed using complex numbers. A phasor is a quantity that contains information about the peak voltage or peak current in a component together with its phase.

Most functions in engineering mathematics are analytic functions, and their study as functions of a complex variable leads to a deeper understanding of their properties and to interrelations in complex that have no analog in real calculus.

I. COMPLEX ARITHMETIC

It was observed early in history that there are equations which are not satisfied by any real number, for example $x^2 + 1 = 0$, $x^2 + 2x + 2 = 0$ etc. This led to the definition of a complex number.

To describe complex numbers, we use a formal symbol j representing $\sqrt{-1}$; then a **complex number** is an expression of the form

$$a + jb \quad a, b \text{ real numbers}$$

If $a = 0$ or $b = 0$, they are omitted (unless both are 0); thus we write

$$a + j0 = a, \quad 0 + jb = jb, \quad 0 + j0 = 0$$

A complex number is a mixture of a real number and an imaginary number. The symbol z is used to denote a complex number.

In the complex number $z = 3 + j5$ the number 3 is called the *real part* of z and is denoted as $\text{Re}(z)$. The number 5 is called the *imaginary part* of z and is denoted $\text{Im}(z)$.

$$\text{So } \text{Re}(3 + j5) = 3 \text{ and } \text{Im}(3 + j5) = 5$$

Notice: the imaginary part of a complex number is a real number so that:

$$\text{Im}(3 + j5) = 5 \text{ and not } j5$$

$$\text{So, a complex number } z = \text{Re}(z) + j\text{Im}(z).$$

Complex numbers have many applications in engineering and science. To use them, we must know how to carry out the usual arithmetical operations.

Let $z_1 = a + jb$ and $z_2 = c + jd$, and making use of the fact that $j^2 = -1$, then

1. Addition and subtraction

$$\begin{aligned} z_1 + z_2 &= a + jb + c + jd \\ &= (a + c) + j(b + d) \end{aligned}$$

$$\begin{aligned} z_1 - z_2 &= a + jb - (c + jd) \\ &= (a - c) + j(b - d) \end{aligned}$$

2. Multiplication

$$\begin{aligned} z_1 \cdot z_2 &= (a + jb)(c + jd) \\ &= ac + jad + jcb + j^2 db \\ &= (ac - bd) + j(ad + bc) \end{aligned}$$

Division is a little more complicated; what is important is not so much the final formula but rather the procedure which produces it; assuming $c + jd \neq 0$, it is:

3. Division

$$\frac{z_1}{z_2} = \frac{a + jb}{c + jd} = \frac{a + jb}{c + jd} \cdot \frac{c - jd}{c - jd} = \frac{ac + bd}{c^2 + d^2} + j \frac{bc - ad}{c^2 + d^2}$$

This division procedure made use of *complex conjugation*: if $z = a + jb$, we define the **complex conjugate** of z to be the complex number

$$\bar{z} = a - jb \quad (\text{note that } z\bar{z} = a^2 + b^2).$$

The product of two conjugate complex numbers is always entirely real.

4. Equal complex numbers

The definition of *equality* between two complex numbers is

$$a + jb = c + jd \Leftrightarrow a = c, b = d$$

5. Modulus of a complex number

The size of a complex number is measured by its **absolute value**, or *modulus*, defined by

$$|z| = |a + jb| = \sqrt{a^2 + b^2} \quad (\text{thus: } z\bar{z} = |z|^2)$$

Examples:

- Express the following complex numbers in standard form:
a) $\frac{4}{j}$ b) $\frac{3+j}{5-j}$
- Find the curve whose equation is $\operatorname{Re}\left(\frac{1}{z}\right) = \frac{1}{4}$
- Show that the real part $\operatorname{Re}(z)$ and imaginary part $\operatorname{Im}(z)$ can be expressed in terms of z and $\bar{z}$.
- Find the real solution of $(4+2j)x + (5-3j)y = 13+j$

Review exercise



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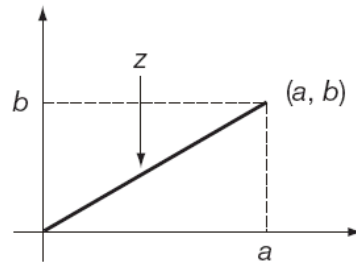
- Simplify (a) j^{12} (b) j^{10} (c) j^{-7}
- Simplify:
(a) $(5-j9) - (2-j6) + (3-j4)$
(b) $(6-j3)(2+j5)(6-j2)$
(c) $(4-j3)^2$
(d) $(5-j4)(5+j4)$
- Multiply $(4-j3)$ by an appropriate factor to give a product that is entirely real. What is the result?
- Simplify $\frac{3-j2}{1+j}$ into the form $a + jb$
- If $\frac{2-a-j5}{1-jb} = 3-j5$ find a and b

Solutions: frame 39 pg 392 Stroud

II. GRAPHICAL REPRESENTATION OF A COMPLEX NUMBER

Argand diagram

The complex number $z = a + jb$ can be represented by the line joining the origin to the point (a, b) set against Cartesian axes.



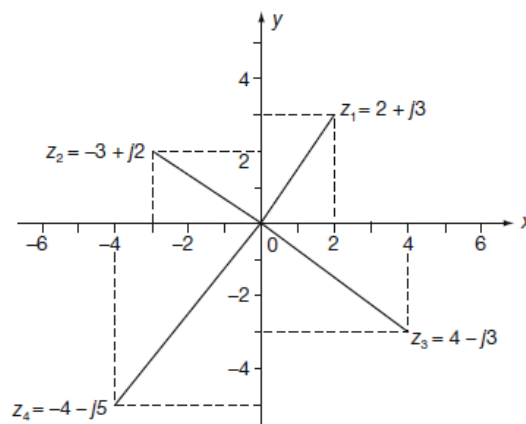
This is called the **Argand** diagram and the plane of points is called the *complex plane*. The two axes are referred to as the real and imaginary axes respectively but note that it is *real numbers* that are plotted on both axes.

Example:

Draw an Argand diagram to represent the complex numbers:

- (a) $z_1 = 2 + j3$ (b) $z_2 = -3 + j2$
 (c) $z_3 = 4 - j3$ (d) $z_4 = -4 - j5$

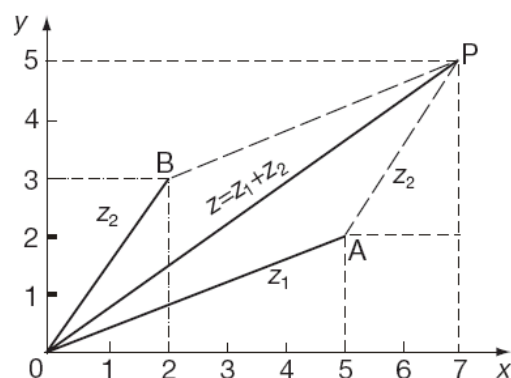
Solution:



Graphical addition/ subtraction of complex numbers

Complex numbers add (subtract) according to the parallelogram rule:

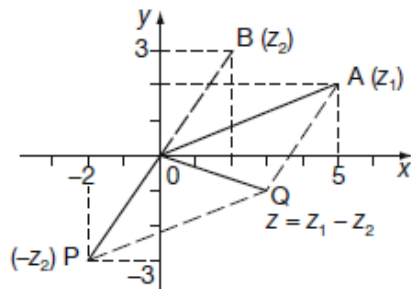
$$(5 + j2) + (2 + j3) = 7 + j5$$



Subtraction is done in a similar way,

$$z_1 - z_2 = z_1 + (-z_2)$$

That is, we draw the lines representing z_1 and the *negative* of z_2 and add them as before. The negative of z_2 is simply a line with the same magnitude (or length) as z_2 but in the opposite direction.



e.g. If $z_1 = 5 + j2$ and $z_2 = 2 + j3$

$$OA = z_1 = 5 + j2$$

$$OP = -z_2 = -(2 + j3)$$

$$\begin{aligned} \text{Then } OQ &= z_1 + (-z_2) \\ &= z_1 - z_2 \end{aligned}$$

III. POLAR FORM OF A COMPLEX NUMBER

It is convenient sometimes to express a complex number $a + jb$ in a different form. On an Argand diagram, let OP be a complex number $a + jb$. Let r = length of the complex number and θ the angle made with the x -axis.

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$$\text{Then } r^2 = a^2 + b^2 \quad r = \sqrt{a^2 + b^2}$$

$$\text{and } \tan \theta = \frac{b}{a} \quad \theta = \tan^{-1} \frac{b}{a}$$

$$\text{Also } a = r \cos \theta \quad \text{and } b = r \sin \theta$$

Since $z = a + jb$, this can be written

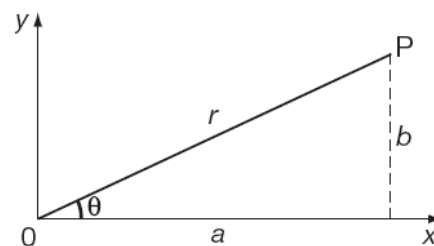
$$z = r \cos \theta + jr \sin \theta$$

$$\text{i.e. } z = r(\cos \theta + j \sin \theta)$$

This is called the *polar form* of the complex number $a + jb$, where

$$r = \sqrt{a^2 + b^2}$$

$$\text{and } \theta = \tan^{-1} \frac{b}{a}$$



When the complex number is written in polar form,

$$r = |z| \quad (\text{absolute value, modulus})$$

We call θ the polar angle or the argument of z . In symbols, one sometimes sees

$$\theta = \arg(z) \quad (\text{polar angle, argument})$$

The absolute value is uniquely determined by $a + jb$, but the polar angle is not, since it can be increased by any integer multiple of 2π . (The complex number 0 has no polar angle). To make θ unique, one can specify

$$0 \leq \theta < 2\pi \quad \text{principal value of the polar angle}$$

This so-called principal value of the angle is sometimes indicated by writing $\text{Arg}(z)$.

$$\text{Then } \arg(z) = \text{Arg}(z) + 2k\pi \quad (\text{where } k = 0, \pm 1, \pm 2, \dots)$$

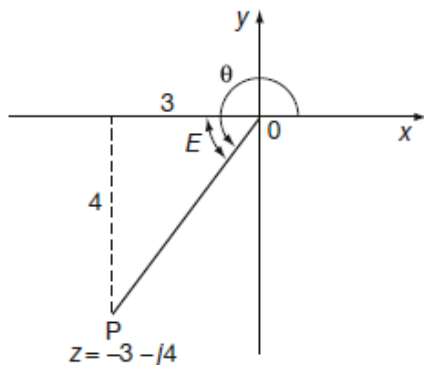
For example, $\text{Arg}(-1) = \pi$, $\arg(-1) = \pm\pi, \pm 3\pi, \pm 5\pi$

Note:

$$\theta = \arg(z) = \frac{\pi}{2} \text{ if } a = 0 \text{ and } b > 0$$

$$\theta = \arg(z) = -\frac{\pi}{2} \text{ or } \frac{3\pi}{2} \text{ if } a = 0 \text{ and } b < 0$$

Warning: In finding θ , there are of course two angles between 0° and 360° , the tangent of which has the value $\frac{b}{a}$. We must be careful to use the angle in the correct quadrant. *Always* draw a sketch of the complex number to ensure you have the right one.



e.g. Find $\arg z$ when $z = -3 - j4$

θ is measured from Ox to OP. We first find E , the equivalent acute angle from the triangle shown:

$$\tan E = \frac{4}{3} = 1.333 \dots \quad \therefore E = 53^\circ 8'$$

Then in this case:

$$\theta = 180^\circ + E = 233^\circ 8' \quad \arg z = 233^\circ 8'$$

Examples:

Express the following complex numbers in polar form

1. $-j$
2. $1 + j$
3. $1 - j$
4. $-1 + j\sqrt{3}$
5. $z = -\sin \frac{\pi}{8} - j \cos \frac{\pi}{8}$
6. $-1 - j\sqrt{3}$
7. Express in the form $a + jb$
 - a) $z = 5(\cos 140^\circ + j \sin 140^\circ)$
 - b) $z = 3(\cos \frac{5\pi}{6} + j \sin \frac{5\pi}{6})$

IV. EXPONENTIAL FORM OF A COMPLEX NUMBER

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There is still another way of expressing a complex number which we must deal with, for it too has its uses. We shall arrive at it this way:

Many functions can be expressed as series. for example,

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots \dots \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} + \dots \dots \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \dots \dots$$

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If we now take the series for e^x and write $j\theta$ in place of x , we get:

$$\begin{aligned} e^{j\theta} &= 1 + j\theta + \frac{(j\theta)^2}{2!} + \frac{(j\theta)^3}{3!} + \frac{(j\theta)^4}{4!} + \dots \dots \dots \\ &= 1 + j\theta + \frac{j^2\theta^2}{2!} + \frac{j^3\theta^3}{3!} + \frac{j^4\theta^4}{4!} + \dots \dots \dots \\ &= 1 + j\theta - \frac{\theta^2}{2!} - \frac{j\theta^3}{3!} + \frac{\theta^4}{4!} + \dots \dots \dots \\ &= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \dots \dots\right) + j\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \dots \dots\right) \\ &= \cos \theta + j \sin \theta \end{aligned}$$

$$\therefore e^{j\theta} = \cos \theta + j \sin \theta \quad \text{Euler's formula}$$

Therefore, $r(\cos \theta + j \sin \theta)$ can now be written as $re^{j\theta}$. This is called the exponential form of the complex number. It is obtained quite easily from the polar form since the r value is the same and the angle θ is the same in both. Note, however that in the exponential form, the angle must be in radians.

The most important reason for polar and exponential representation is that multiplication and division of complex numbers is particularly simple when they are written in these forms. Indeed, by using Euler's formula and the trigonometric addition formulas, it is not hard to show

$$e^{j\theta_1} \cdot e^{j\theta_2} = e^{j(\theta_1 + \theta_2)}$$

Thus we see that the complex exponentials follow the same exponential addition law as the real exponential. This leads to the following simple rules for multiplying and dividing complex numbers written in exponential form.

$$\text{Given } z_1 = r_1 e^{j\theta_1} \text{ and } z_2 = r_2 e^{j\theta_2}$$

- **multiplication rule** $z_1 \cdot z_2 = r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 r_2 e^{j(\theta_1 + \theta_2)}$

to multiply two complex numbers, you multiply the absolute values and add the angles.

- **reciprocal rule** $\frac{1}{re^{j\theta}} = \frac{1}{r}e^{-j\theta}$
- **division rule** $\frac{z_1}{z_2} = \frac{r_1 e^{j\theta_1}}{r_2 e^{j\theta_2}} = \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)}$

to divide by a complex numbers, divide by its absolute value and subtract its angle.

Similarly multiplication and division is easier in polar form:

The polar form at first sight seems to be a complicated way of representing a complex number. However it is very useful, as we shall see. Suppose we multiply together two complex numbers in this form:

$$\text{Let } z_1 = r_1(\cos \theta_1 + j \sin \theta_1) \text{ and } z_2 = r_2(\cos \theta_2 + j \sin \theta_2)$$

$$\text{Then } z_1 z_2 = r_1(\cos \theta_1 + j \sin \theta_1)r_2(\cos \theta_2 + j \sin \theta_2)$$

$$= r_1 r_2 (\cos \theta_1 \cos \theta_2 + j \sin \theta_1 \cos \theta_2 + j \cos \theta_1 \sin \theta_2 + j^2 \sin \theta_1 \sin \theta_2)$$

Rearranging the terms and remembering that $j^2 = -1$, we get

$$z_1 z_2 = r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + j(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)]$$

$$\begin{aligned} \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 &= \cos(\theta_1 + \theta_2) \\ \sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2 &= \sin(\theta_1 + \theta_2) \end{aligned}$$

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In that case, $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2)]$.

Note this important result. We have just shown that

$$r_1(\cos \theta_1 + j \sin \theta_1) \cdot r_2(\cos \theta_2 + j \sin \theta_2) = r_1 r_2 [\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2)]$$

i.e. To multiply together two complex numbers in polar form,

- multiply the r 's together,
- add the angles, θ , together.

The division rule for polar form can be proven the same way.

So we now have two important rules:

If $z_1 = r_1(\cos \theta_1 + j \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + j \sin \theta_2)$

$$\text{then (a) } z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + j \sin(\theta_1 + \theta_2)]$$

$$\text{and (b) } \frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + j \sin(\theta_1 - \theta_2)]$$

Examples:

1. Change to exponential form:

$$\text{a) } -1 + j \qquad \text{b) } \sqrt{3} - j \qquad \text{c) } \frac{1 + j\sqrt{3}}{\sqrt{3} + j}$$

Write each of your answers in a), b) and c) in polar form i.e. $r(\cos \theta + j \sin \theta)$

- Simplify
 - $2(\cos 120^\circ + j \sin 120^\circ) \times 4(\cos 20^\circ + j \sin 20^\circ)$
 - $a(\cos \theta + j \sin \theta) \times b(\cos \phi + j \sin \phi)$

3. If $z_1 = 8(\cos 65^\circ + j \sin 65^\circ)$ and $z_2 = 4(\cos 23^\circ + j \sin 23^\circ)$ then

(a) $z_1 z_2 = \dots\dots\dots$

and (b) $\frac{z_1}{z_2} = \dots\dots\dots$

4. Simplify: $\frac{5(\cos 60^\circ + j \sin 60^\circ) \times 4(\cos 30^\circ + j \sin 30^\circ)}{2(\cos 50^\circ + j \sin 50^\circ)}$

21 Review exercise



Work all these questions and then turn on to Frame 22 and check your results.

- 1 Express in polar form, $z = -4 + j2$.
- 2 Express in true polar form, $z = 5(\cos 55^\circ - j \sin 55^\circ)$.
- 3 Simplify the following, giving the results in polar form:
 - (a) $3(\cos 143^\circ + j \sin 143^\circ) \times 4(\cos 57^\circ + j \sin 57^\circ)$
 - (b) $\frac{10(\cos 126^\circ + j \sin 126^\circ)}{2(\cos 72^\circ + j \sin 72^\circ)}$
- 4 Express in the form $a + jb$:
 - (a) $2(\cos 30^\circ + j \sin 30^\circ)$
 - (b) $5(\cos 57^\circ - j \sin 57^\circ)$

Solutions: frame 22 pg 414 Stroud

V. ROOTS OF A COMPLEX NUMBER

To solve linear differential equations with constant coefficients, you need to be able to find the real and complex roots of polynomial equations. Though a lot of this is done today with calculators and computers, one still has to know how to do an important special case by hand: finding the roots of

$$z^n = \alpha$$

Where α is a complex number, i.e finding the n -th roots of α .

DeMoivre's Theorem

The very important formula

$$[r(\cos \theta + j \sin \theta)]^n = r^n (\cos n\theta + j \sin n\theta)$$

is known as *DeMoivre's Theorem*.

It says that to raise a complex number in polar form to any power n , we raise r to the power n and multiply the angle by n .

For the derivation of this theorem, see frames 24-25 pg 415 Stroud.

$$\begin{aligned}\text{e.g. } [4(\cos 50^\circ + j \sin 50^\circ)]^2 &= 4^2[\cos(2 \times 50^\circ) + j \sin(2 \times 50^\circ)] \\ &= 16(\cos 100^\circ + j \sin 100^\circ) \\ \text{and } [3(\cos 110^\circ + j \sin 110^\circ)]^3 &= 27(\cos 330^\circ + j \sin 330^\circ)\end{aligned}$$

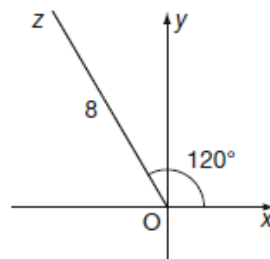
This is where the polar form really comes into its own! For DeMoivre's theorem also applies when we are raising the complex number to a fractional power, i.e. when we are finding the roots of a complex number.

e.g. To find the square root of $z = 4(\cos 70^\circ + j \sin 70^\circ)$

$$\begin{aligned}\text{We have } \sqrt{z} = z^{\frac{1}{2}} &= [4(\cos 70^\circ + j \sin 70^\circ)]^{\frac{1}{2}} \quad \text{i.e. } n = \frac{1}{2} \\ &= 4^{\frac{1}{2}} \left(\cos \frac{70^\circ}{2} + j \sin \frac{70^\circ}{2} \right) \\ &= 2(\cos 35^\circ + j \sin 35^\circ)\end{aligned}$$

It works every time, no matter whether the power is positive, negative, whole number or fraction.

Look again at finding a root of a complex number. Let us find the cube root of $z = 8(\cos 120^\circ + j \sin 120^\circ)$. Here is the given complex number shown on an Argand diagram:



Of course, we could say that θ was '1 revolution +120°': the vector would still be in the same position, or (2 revs +120°), (3 revs +120°), etc

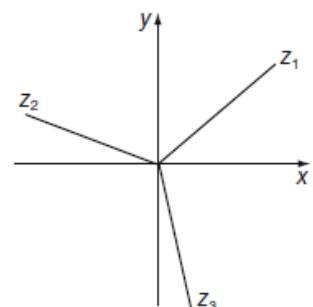
$$\begin{aligned}z &= 8(\cos 120^\circ + j \sin 120^\circ) \\ \text{i.e. or } z &= 8(\cos 480^\circ + j \sin 480^\circ) \\ \text{or } z &= 8(\cos 840^\circ + j \sin 840^\circ) \quad \text{etc}\end{aligned}$$

and if we now apply DeMoivre's theorem to each of these, we get

$$\begin{aligned}z^{\frac{1}{3}} &= 8^{\frac{1}{3}} \left(\cos \frac{120^\circ}{3} + j \sin \frac{120^\circ}{3} \right) = 2(\cos 40^\circ + j \sin 40^\circ) \\ &= 8^{\frac{1}{3}} \left(\cos \frac{480^\circ}{3} + j \sin \frac{480^\circ}{3} \right) = 2(\cos 160^\circ + j \sin 160^\circ) \\ &= 8^{\frac{1}{3}} \left(\cos \frac{840^\circ}{3} + j \sin \frac{840^\circ}{3} \right) = 2(\cos 280^\circ + j \sin 280^\circ) \quad \text{etc}\end{aligned}$$

If we put each of these on an Argand diagram, we see we have three quite different results for the cube root of z and also that the fourth diagram would be a repetition of the first. Any subsequent multiples of 360° or 2π , merely repeat these three positions.

We see therefore, that there are 3 cube roots of a complex number. Also, if you consider the angles, you see that the 3 roots are equally spaced round the diagram, separated by 120° .



We find any roots of a complex number in the same way:

- (a) Apply DeMoivre's theorem to find the first of the n roots.
- (b) The other roots will then be distributed round the diagram at regular intervals of $\frac{360^\circ}{n}$.

A complex number, therefore, has:

- 2 square roots, separated by $\frac{360^\circ}{2}$ i.e. 180°
- 3 cube roots, separated by $\frac{360^\circ}{3}$ i.e. 120°
- 4 fourth roots, separated by $\frac{360^\circ}{4}$ i.e. 90° etc.

In general,

$$\sqrt[n]{r}(\cos \theta + j \sin \theta) = \sqrt[n]{r} \left[\cos \frac{\theta + 2\pi k}{n} + j \sin \frac{\theta + 2\pi k}{n} \right]$$

where $k = 0, 1, 2, \dots, n-1$

And

$$\sqrt[n]{r}e^{j\theta} = \sqrt[n]{r}e^{j(\theta + 2\pi k)/n}$$

where $k = 0, 1, 2, \dots, n-1$

Examples:

1. Find in Cartesian form all values of a) $\sqrt[3]{1}$ b) $\sqrt[4]{j}$
2. Solve the equation $z^5 + 32 = 0$
3. Solve the equation $x^6 - 2x^3 + 2 = 0$

Test exercise 2



- 1 Express in polar form, $z = -5 - j3$.

- 2 Express in the form $a + jb$:

(a) $2\sqrt[5]{156^\circ}$

(b) $5\sqrt[3]{37^\circ}$



- 3 If $z_1 = 12(\cos 125^\circ + j \sin 125^\circ)$ and $z_2 = 3(\cos 72^\circ + j \sin 72^\circ)$, find (a) $z_1 z_2$ and (b) $\frac{z_1}{z_2}$ giving the results in polar form.

- 4 If $z = 2(\cos 25^\circ + j \sin 25^\circ)$, find z^3 in polar form.



- 5 Find the three cube roots of $8(\cos 264^\circ + j \sin 264^\circ)$ and state which of them is the principal cube root. Show all three roots on an Argand diagram.

Answers on pg 1137 Stroud